RILSAMID®
AMNO TLD

PA12, MHLR, 12-010

Rilsamid® AMNO TLD resin is a natural polyamide. This grade is designed for injection molding.

PROPERTIES	DRY / COND	UNIT	TEST STANDARD
RHEOLOGICAL PROPERTIES			
Melt Volume-Flow Rate	57 / *	cm³/10 min	ISO 1133
Temperature	235 / *	°C	-
	455 / *	°F	
Load	2.16 / *	kg	-
	4.76 / *	lb	
MECHANICAL PROPERTIES			
Tensile Modulus	1450 / 1170	MPa	ISO 527-1/-2
	210000 / 170000	psi	
Yield Stress	42 / 38	MPa	ISO 527-1/-2
	6090 / 5510	psi	
Yield Strain	7 / 7	%	ISO 527-1/-2
Nominal Strain at Break	>50 / >50	%	ISO 527-1/-2
Charpy Impact Strength, +23°C	- / No Break	kJ/m²	ISO 179/1eU
Charpy Impact Strength, -30°C	- / No Break	kJ/m²	ISO 179/1eU
Charpy Notched Impact Strength, +23°C	- / 9	kJ/m²	ISO 179/1eA
	- / 4.28	ftlb/in²	
Charpy Notched Impact Strength, -30°C	- / 5	kJ/m²	ISO 179/1eA
	- / 2.38	ftlb/in²	
THERMAL PROPERTIES			
Melting Temperature, 10°C/min	178 / *	°C	ISO 11357-1/-3
Temp. of Deflection Under Load, 1.80 MPa	55 / *	°C	ISO 75-1/-2
	131 / *	°F	
Temp. of Deflection Under Load, 0.45 MPa	135 / *	°C	ISO 75-1/-2
	275 / *	°F	
Vicat Softening Temperature, 50°C/h 50N	142 / *	°C	ISO 306
	288 / *	°F	
Coeff. of Linear Thermal Expansion, parallel	130 / *	E-6/K	ISO 11359-1/-2
Coeff. of Linear Thermal Expansion, normal	120 / *	E-6/K	ISO 11359-1/-2
Burning Behav. at 1.5 mm Nominal Thickness	HB / *	class	IEC 60695-11-10
Burning Behav. at Thickness h	V-2 / *	class	IEC 60695-11-10
Thickness Tested	3.2 / *	mm	-
	0.1260 / *	in	

Arkema France - A French "société anonyme", registered in the Nanterre (France) Trade and Companies Register under the number 319 632 790 SDC/11-2018
Source: automatically generated TDS from Material Database on 12-08-2024

RILSAMID®
AMNO TLD

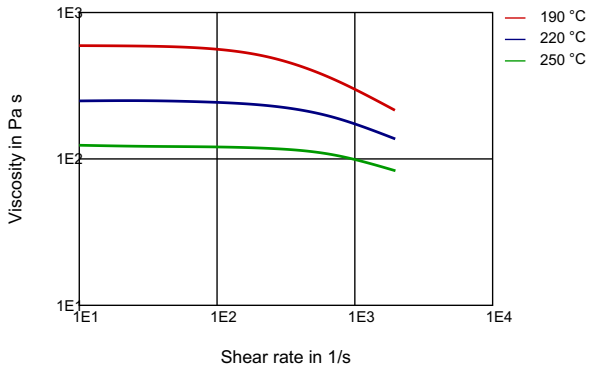
Yellow Card available	yes / *	-	-
Oxygen Index	22 / *	%	ISO 4589-1/-2
ELECTRICAL PROPERTIES			
Relative Permittivity, 100Hz	4 / -	-	IEC 60250
Relative Permittivity, 1MHz	3 / -	-	IEC 60250
Dissipation Factor, 100Hz	779 / -	E-4	IEC 60250
Dissipation Factor, 1MHz	290 / -	E-4	IEC 60250
Volume Resistivity	- / 1E12	Ohm*m	IEC 62631-3-1
Surface Resistivity	* / 1E14	Ohm	IEC 62631-3-2
Dielectric (Electric) Strength	- / 30	kV/mm	IEC 60243-1
	- / 762	kV/in	
Comparative Tracking Index	* / 600	-	IEC 60112
OTHER PROPERTIES			
Water Absorption, 23°C, immersion, equilibrium	1.8 / *	%	ISO 62
Humidity Absorption, 23°C, RH50%, equilibrium	0.7 / *	%	ISO 62
Density	1020 / 1020	kg/m³	ISO 1183
	1.02 / 1.02	g/cm³	

Arkema France - A French "société anonyme", registered in the Nanterre (France) Trade and Companies Register under the number 319 632 790 SDC/11-2018
Source: automatically generated TDS from Material Database on 12-08-2024

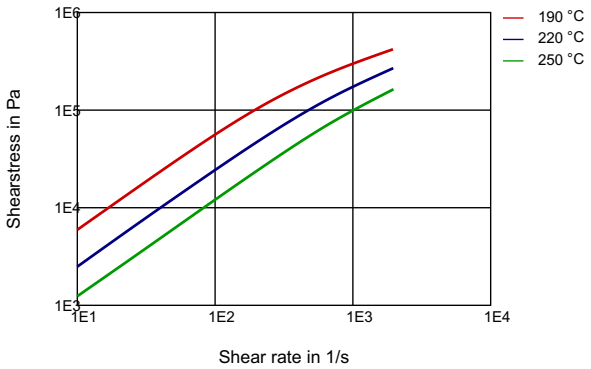
RILSAMID®
AMNO TLD

DIAGRAMS

VISCOSITY-SHEAR RATE



SHEARSTRESS-SHEAR RATE



Processing conditions, Injection:

- Typical melt temperature (Min / Recommended / Max) : 230°C / 270°C / 290°C.
- Mold temperature : 20 - 40°C
- Drying time and temperature (only necessary for bags opened for more than two hours) : 4-6 hours at 80 - 90°C.

PROCESSING

Injection Molding

DELIVERY FORM

Pellets

ADDITIVES

Release agent

SPECIAL CHARACTERISTICS

Heat Stabilized, Light Stabilized

REGIONAL AVAILABILITY

North America, Europe, Asia Pacific, South and Central America, Near East/Africa

Headquarters:

Arkema France
420 rue d'Estienne d'Orves
92705 Colombes Cedex France
T +33 (0)1 49 00 80 80
hpp.arkema.com

Arkema Inc. – High Performance Polymers

900 First Avenue
King of Prussia, PA 19406
Tel.: +1 610 205 7000
hpp.arkema.com

Arkema France - A French "société anonyme", registered in the Nanterre (France) Trade and Companies Register under the number 319 632 790 SDC/11-2018
Source: automatically generated TDS from Material Database on 12-08-2024

RILSAMID®
AMNO TLD

The statements, technical information and recommendations contained herein are believed to be accurate as of the date hereof. Since the conditions and methods of use of the product and of the information referred to herein are beyond our control, ARKEMA expressly disclaims any and all liability as to any results obtained or arising from any use of the product or reliance on such information; NO WARRANTY OF FITNESS FOR ANY PARTICULAR PURPOSE, WARRANTY OF MERCHANTABILITY OR ANY OTHER WARRANTY, EXPRESSED OR IMPLIED, IS MADE CONCERNING THE GOODS DESCRIBED OR THE INFORMATION PROVIDED HEREIN. The information provided herein relates only to the specific product designated and may not be applicable when such product is used in combination with other materials or in any process. The user should thoroughly test any application before commercialization. Nothing contained herein constitutes a license to practice under any patent and it should not be construed as an inducement to infringe any patent and the user is advised to take appropriate steps to be sure that any proposed use of the product will not result in patent infringement.

Arkema France - A French "société anonyme", registered in the Nanterre (France) Trade and Companies Register under the number 319 632 790 SDC/11-2018
Source: automatically generated TDS from Material Database on 12-08-2024